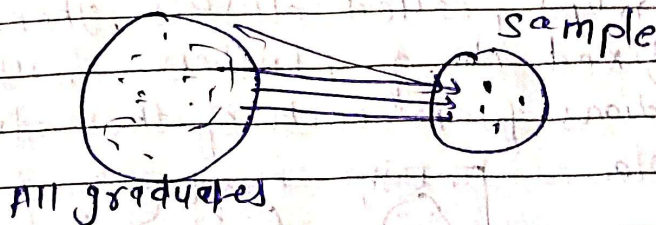


Importance of Statistical Analysis

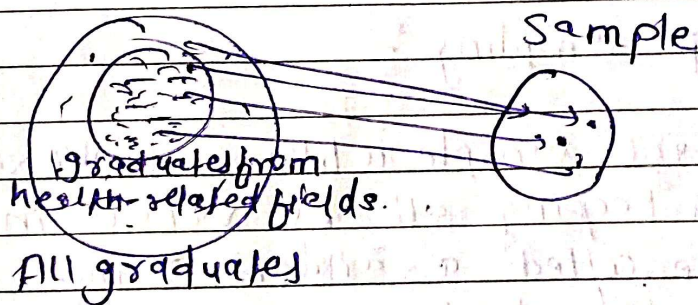
→ Statistics allows us to use the proper methods for collecting data, employ the correct Analysis & effectively present the results.

1) Sampling Principles & Sampling Strategies

i) Sampling from population:-



2) Biased sampling & its causes :-



II Observational studies

1. 4 Types of sampling

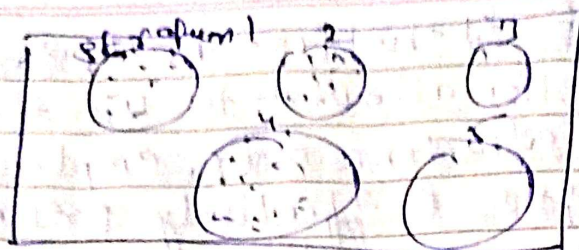
→ i) Simple random sampling



→ Randomly select cases from population, No implied connections b/w the points that are selected.

2) Stratified Random sampling

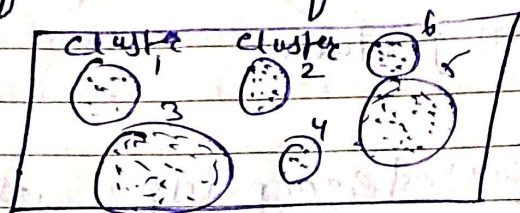
→ divide & conquer strategy. Population is divided into groups called strata.



Strata are made up of similar observations. We take a simple random sample from each stratum.

3) Cluster sampling:-

→ In a cluster sample, we break up the population into many groups, called clusters, then we sample a fixed no. of clusters & include all observations from each of those clusters in the sample.



4) Multi stage sampling:-

A Multistage sample is like a cluster sample, but rather than keeping all observations in each cluster, we collect a random sample within each selected cluster.

We randomly select a subset of each cluster to be included in the sample rather than measuring every case in each sampled cluster.

